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1.Which of the following is true about 1NF?

Answer: (c) All attributes must be atomic

Explanation: 1NF requires every attribute to contain only single, indivisible values.

2.A relation is in 2NF if it is in 1NF and:

Answer: (b) Has no partial dependencies on primary key

Explanation: 2NF removes partial dependency of non-key attributes on part of a composite primary key.

3.Which normal form deals with multi-valued dependencies?

Answer: (d) 4NF

Explanation: Fourth Normal Form (4NF) specifically eliminates non-trivial multivalued dependencies.

4.BCNF is a stricter version of:

Answer: (c) 3NF

Explanation: BCNF is stronger than 3NF because every determinant must be a candidate key.

5.Functional dependency X → Y means:

Answer: (b) X determines Y

Explanation: If two tuples have the same value of X, they must have the same value of Y.

6.Which of the following anomaly is NOT associated with unnormalized relations?

Answer: (d) Retrieval anomaly

Explanation: The common anomalies are insertion, deletion, and update anomalies.

7.A superkey is:

Answer: (b) Any set of attributes that uniquely identifies tuples

Explanation: A superkey uniquely identifies each tuple. It may contain extra attributes, unlike a candidate key.

8.If R(A,B,C) has FD: A → B and B → C, then C is:

Answer: (c) Transitively dependent on A

Explanation: Since A → B and B → C, C depends on A through B. Therefore, it is a transitive dependency.

9.5NF is also known as:

Answer: (c) Project-Join Normal Form

Explanation: Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJ/NF).

10.Lossless-join decomposition ensures:

Answer: (a) No data is lost during decomposition

Explanation: A lossless decomposition allows the original relation to be reconstructed by joining the decomposed relations.